

Energy poverty trends and determinants in Pakistan: Empirical evidence from eight waves of HIES 1998–2019

Ashar Awan^{a,b}, Faik Bilgili^c, Dil Bahadur Rahut, PhD^{d,*}

^a FEAS, Economics, Erciyes University, Melikgazi, Kayseri, Turkey

^b Nisantasi University Graduate School, Istanbul, Turkey

^c FEAS, Economics, Erciyes University, Turkey

^d Asian Development Bank Institute, Kasumigaseki Building 8F, 3-2-5 Kasumigaseki, Chiyoda-ku, Tokyo, 100-6008, Japan

ARTICLE INFO

Keywords:

Energy poverty
Assets
Pakistan
Education
HIES
Ten percent
Double median approach
Affordability
Energy social impacts
Inclusion

ABSTRACT

United Nations Sustainable Development Goals (SDGs) have incorporated clean and affordable fuel access as the 7th goal because energy is crucial for economic development and well-being. However, the increase in energy prices, volatility, removal of subsidies, fiscal burden and limited access to modern fuel, and financial burden on the households have led to an increase in energy poverty. Although energy poverty is widely prevalent worldwide, it is more pervasive in developing countries. Yet, there is limited research on energy poverty in developing countries, and the government policy and program do not focus on energy poverty. Therefore, this study examines the energy poverty trend and factors influencing energy poverty in Pakistan using eight rounds of nationally representative Household Income and Expenditure (HIES) survey data collected during 1998–2019. This research adopted two different measures of energy poverty, namely ten percent and double median. The result shows that despite significant economic progress, energy poverty has increased during the last two decades. The study further investigated the determinants of energy poverty in Pakistan using the Probit model, and the results revealed that households with a low endowment, less education, and female-headed families are more likely to be energy poor. Given Pakistan's current energy poverty situation, it is challenging to meet SDG 7 of providing clean and affordable energy access to all by 2030. Therefore, sizeable clean energy programs targeting the poor with low education, families living in rural areas, and female-headed households are needed.

1. Introduction

All the phases of human history are reliant on any known form of energy for survival and prosperity [1]. The importance of energy for livelihood, well-being, and prosperity is evident from the continuous rise in per capita energy consumption, particularly during the last century [2]. Along with industrial and transport sectors, energy is crucial for cooking, heating, and lighting for the residential sector. Therefore, ensuring access to affordable energy is the priority of governments in developed and developing countries. Lack of access to clean and affordable energy combined with low income or economic status pushes households into energy poverty.

Recognizing the importance of energy poverty for sustainable development and well-being, the UN General Assembly marked 2012 as the “International year of sustainable energy for all” and affirmed its

commitment to promote affordable energy access to improve socio-economic well-being [3]. In addition, energy poverty is amongst the United Nations Sustainable Development Goals (UN-SDGs) goals. Goal seven of UN-SDGs mainly targets energy poverty through “ensuring access to affordable, reliable, sustainable and modern energy for all”.

At the macro level, energy poverty increases income poverty¹ [4,5] and reduces the social and economic progress of societies [6–8]. At the micro-level, energy poverty is a barrier to education, women empowerment, and the health and well-being of the household members [9]. Furthermore, reducing energy poverty helps mitigate outdoor and indoor air pollution [10–13]. Access to clean sources of domestic fuel has several benefits such as better environment, human health, well-being, particularly of female members [14]. The use of solid fuel for cooking adversely affects human health [11,15–21]. Evidence from developing countries shows that energy poverty is found to cause indoor air pollution and, consequently, a reason for premature death among children

* Corresponding author.

E-mail addresses: asharawan786@hotmail.com (A. Awan), fbilgili@erciyes.edu.tr (F. Bilgili), drahut@adbi.org (D.B. Rahut).

¹ For a detailed account of income poverty, see [127].

List of abbreviations

SDGs	Sustainable Development Goals
HIES	Household Integrated Economic Survey
PSLM	Pakistan Social and Living Measure
DHS	Demographic Health Survey
MEPI	Multidimensional Energy Poverty Index
GHG	Green House Gases
IEA	International Energy Agency
MIS	Minimum Income Standard
AFCP	After Fuel Cost Poverty
RIA	Residual Income Approach
PSU	Primary Sampling Units
PCA	Principal Component Analysis

[22–25]. Whereas the use of dirty fuel such as firewood is not only harmful to human health but also detrimental to the natural environment [26]. Firewood usage for cooking and heating by households contributes to deforestation in developing countries. Finally, the use of clean energy enhances the well-being and economic conditions of the households [13,27–31].

Energy poverty (fuel poverty) has been gaining attention from researchers and development professionals [22,32–35] since the seminal study of Bradshaw and Hutton [36]. Energy poverty originates from low income, lack of access to clean fuel [22,35], or high energy prices [37]. The concept of energy poverty is not only about low income, but it also includes the lack of ability to fulfill some basic and crucial needs [38]. Recently, several empirical studies focused on energy poverty [39–44]; however, the complexity of defining energy poverty still lacks consensus. In addition, the measurement of energy poverty is still a challenge [45,46]. The overlapping concepts of energy poverty [47], fuel poverty [12], energy vulnerability [48] energy affordability [49] try to explain a situation in which a household faces scarcity of energy services [31,44].

In developing countries, such as Pakistan, achieving SDG goal 7 requires a clear picture of two components of this goal, namely, ‘access and ‘affordability’ to clean, reliable, modern, and sustainable energy. First is related to access and preference for clean fuel such as gas and electricity. This aspect of energy poverty deals with the access and clean fuel usage that is discussed in previous energy literature on Pakistan [50, 51]. On the other hand, affordability is related to income and expenditure and the financial burden caused by energy-related expenses. Keeping in mind the importance of affordability, the present study mainly focuses on energy poverty related to income and expenditure relationships and ignores fuel choice and energy access issues. The goal seven of SDGs, which targets energy poverty in all forms, including access and affordability, requires household level analysis in Pakistan based on income and expenditure surveys. Therefore, this study examines the energy poverty situation in Pakistan by using nationally representative data of HIES surveys from 1998 to 2019 and covering 142,537 households.

2. The energy poverty problem in Pakistan

Pakistan started its power sector reforms process in the last decade of the previous century. These reforms include privatization and unbundling of power sector. Moreover, in 2008 and 2013, Pakistan received loans from International Monetary Fund (IMF) with a promise to reduce subsidies and increase tariffs. However, there is a tradeoff between implementing these reforms and achieving goal seven of UNSDGs [52]. In February 2016, the National Assembly of Pakistan unanimously adopted Sustainable Development Goals (SDGs) as the national agenda [53], which includes universal access to affordable and clean energy as

one of the targets to be achieved by 2030 [54]. Given the competing priorities and limited available resources, it is challenging to achieve this goal of universal access to affordable and clean energy.

Energy poverty is prevalent in developing countries like Pakistan [55], however, it is relatively less explored [56]. Similar to other developing countries, Pakistan prioritizes income poverty, but, energy poverty receives less attention [50]. Mirza and Kemp [57] find that in rural Punjab (Pakistan), many households rely on traditional fuel such as fuelwood, dung cake, etc., and the remoteness and occupation of the head rather than the income influences energy poverty. Using Pakistan Social and Living Measure (PSLM) 2007–08 and Multidimensional Energy Poverty Index (MEPI), Sher et al. [58] found that 29% of urban households and 71% of the rural households are energy poor. A recent study on energy poverty in Pakistan uses PSLM data 2014–15 and found that 55% of households are in multidimensional energy poverty [50]. Nussbaumer et al. [59] calculated the intensity of energy poverty and Multidimensional Energy Poverty Index (MEPI) for Pakistan using Demographic Health Survey (DHS) data 2006–07 and found head-count-income-poverty and MEPI scores as 0.65 and 0.45, respectively. In another study, Rehmat and Mirza [51] analyzed energy poverty by employing the logit model to find determinants of MEPI based energy poverty in Pakistan. This study used PSLM 2014–15 data for Pakistan. Mahmood and Shah [60] used PSLM 2008–09 and 2011–12 data and found that rural households were more likely to be energy poor than urban households, and the MEPI score declined from 0.45 based on 2008–09 data to 0.264 in 2011–12. Recently, Haleem et al. [61] applied MEPI to calculate energy poverty for Pakistan using PSLM 2013–14 data and found that 70% of households lack clean fuel for cooking and 47% for heating.

Existing literature on energy poverty is based on one-year [62] or at the most two-year data [60] and mostly uses old data. Therefore, it is crucial to use the most recent data to evaluate and understand the energy poverty in Pakistan as things have changed significantly during the last decades. In addition, it is of utmost importance to examine the issue of energy poverty using large data sets covering a longer period to understand the trend and examine the robustness of the finding. Furthermore, Pakistan is a country with various languages, food diversity, weather, and social settings, therefore, a national-level study is essential to understand region disparities and differences between dominant and marginalized groups. Thus, against this backdrop, the present study examines energy poverty using the most recent data and eight waves of Household Income and Expenditure Survey of Pakistan (HIES) from 1998 to 2019.

This study contributes to the existing literature in five ways. Firstly, this study provides estimates of energy poverty for Pakistan using two different measures-namely ten percent and double median, which allow us to comprehensively understand energy poverty and compare the finding from different methods. Secondly, it provides expenditure-based estimates for energy poverty which is missing in energy poverty literature on Pakistan. Thirdly it provides a comprehensive data analysis using recent data from eight rounds of Household Income and Expenditure Survey (HIES) covering the last two decades, yet unexplored data by literature on energy poverty. Fourthly, this is one of the few studies that use eight waves of nationally representative household data covering two decades for 142,537 households. Finally, it makes an extensive review of literature on energy poverty and different energy poverty measures, which is generally lacking.

The remaining part of the study is as follows; Section 3 presents an in-depth literature review about definitions, measurement, and determinants of energy poverty. It is followed by Section 4, which provides details about the data and methodology involved in this study. Section 5 presents the result and discussion. Finally, Section 6 concludes with a policy recommendation.

3. Literature review

Reducing energy poverty improves the health and well-being of an individual household and helps reduce GHG emissions as energy sectors contribute to 73% of the total GHG emission. The present study adds to the literature on energy poverty available for Pakistan [60], Bangladesh [63], USA [39], Turkey [64], Poland [65], Vietnam [66], Japan [67], Greece [68], India [69], China [44]. Available literature applied different indicators to measure energy poverty. Various definitions for energy poverty are proposed in previous literature to cater to the requirements of different geographic regions and socio-economic conditions [13].

3.1. Defining energy poverty?

The definition of energy poverty barely received consensus in the literature. Energy poverty is defined as a situation in which families are not able to affordable energy for heating or cooling purposes and fulfill their energy needs [70]. Usually, energy poverty is used to express all forms of energy poverty [22]. Energy poverty incorporates everything associated with the energy-related concept of poverty [71]. According to the IEA, energy poverty is defined as a lack of access to commercial and clean fuels, electricity, efficient equipment, and high dependence on biomass, causing pollution [72]. Energy Poverty Observatory 2021 defines energy poverty as the absence of vital energy services to individuals and households. However, Winkler et al. [73] defined energy poverty as a state in which households spend a large share of income on energy-related expenditures. The UK government adopted the definition by Hills, which defines energy poverty as “when the cost of basic living needs in energy is higher than the social average, and the residual income is lower than the official economic poverty line” [12,74]. Another study defined energy poverty as “when the amount of energy used is lower than the minimum need to sustain a livelihood” [63]. On the other hand, Gonzalez-Eguino [46] defined energy poverty on the basis of enough energy consumption to meet basic needs. Likewise, on the basis of affordability EU Energy Observatory described energy poverty as a situation in which sufficient energy to meet domestic needs is unaffordable [75]. In short, affordability is a complex phenomenon to define, conceptualize, and empirically operationalize [63,76].

According to Amartya Sen, poverty measurement has two steps [77, 78]. The first step is the identification of the poor, and the second is aggregation. Identification is possible by utilizing a poverty line, while aggregation needs some common poverty characteristics among various people that can serve as a single measure [79]. In this context, Day et al. [11] define energy poverty as a “situation of inability to realize the essential capabilities as a result of insufficient access to affordable, reliable and safe energy services, and taking into account the alternative means of realizing these capabilities reasonably”. As the circumstances vary from region to region, and no official definition of energy poverty is available, there are no universally applied methods to determine the threshold of energy poverty. Therefore, researchers tend to employ different indicators to construct the energy poverty measures.

3.2. Measuring energy poverty

Energy poverty measurement depends on the regional characteristics, weather conditions, and living standards of the study area. Energy is not like nutrition, where a minimum standard can be set scientifically as a benchmark and serve as a threshold point [80]. Furthermore, energy poverty benchmark criteria need continuous updates over time due to changes in technology. An early measure of energy poverty in literature is the famous ten percent approach proposed by Boardman [81], and it has been adopted in several recent research [68,82–87]. According to this method, if a household spends more than 10% of its income or total expenditures on fuel/energy, it is classified as energy poor. Although the 10% measure of energy poverty is very straightforward and easy to

interpret, it possesses certain limitations due to: (a) high sensitivity to fuel prices; (b) under and overestimating the magnitude of the energy poverty when fuel prices are very high or very low; (c) the setting of a subjective threshold at 10% is arbitrary; finally; (d) no linkage with household income [88].

Another measure of energy poverty in earlier literature is the double median method. According to this method, a household is energy poor if its fuel cost is more than double the national median expenditures on fuel. The use of median rather than mean reflects the importance of relative poverty [89]. The concept of the median is an older concept than the ten percent measure of energy poverty [90]. The twice median concept has been used elsewhere in the literature, for instance, in medicine [91], transport [92], and economics [93,94]. The advantage of using the median is that it divides the sample into two groups and keeps estimation insensitive to extreme scores, and it is appreciated internationally for income and expenditure analysis [95].

Besides single indicators to measure energy poverty, various studies used more than one indices [10,33,44,67,96,97]. Recently, Nussbaumer et al. [98] offered multidimensional energy poverty indices to capture comprehensive measures of energy poverty. In the multidimensional approach, energy poverty is estimated from six weighted indicators covering information about clean cooking fuel, electricity availability, home appliances, municipal pollution, entertainment, communication, and educational devices. Some researchers also devised their methods to measure energy poverty based on their requirements and specific conditions [3,44,67]. Moore [99] developed Minimum Income Standard (MIS) [100], which is suitable for evaluating energy equity-linked with the energy poverty issue. According to this indicator, a household is energy poor if, after paying housing costs, the remaining or residual income is less to support energy costs. Some other studies used these indicators with a slight alteration in the mechanism of its calculation [99,101]. The MIS indicator is a good measure of energy poverty as it can be replicated in other EU countries with a small change in minimum income standard. MIS is mainly based on objective measures; however, subjectivity is involved in setting minimum income standards [102].

The other available indicator is After Fuel Cost Poverty (AFCP) [103] which classifies a household as energy-poor, if after paying for housing and fuel expenditure, the remaining income is less than the minimum acceptable income [102]. Residual Income Approach (RIA) classifies a household as energy poor if, after paying bills for gas and electricity, it does not have sufficient financial resources to afford the minimum required consumption of other goods [43]. RIA considers the poverty line as the standard of minimum residual income required to afford necessities.

In summary, while several approaches are available to estimate energy poverty, the financial burden due to energy expenditures is still a problem that is well recognized. Moreover, in Pakistan, energy-related expenditure burden and energy poverty is receiving attention due to rising energy prices. It is argued that a uniform methodology to assess energy poverty in developing countries is required [50], but it is far from reality due to the lack of a uniform dataset. If the study aims at analyzing consumption and other economic variables to gauge energy poverty, multidimensional indices such as MEPI face certain limitations. However, ten percent and double median indicators are reliable due to their less requirement of multidimensional data [42,49,79,104,105]. Therefore, the present study utilized two energy poverty measures: ten percent and double median approaches.

3.3. Factor influencing energy poverty

Earlier literature on energy poverty documented several factors that influence energy poverty. These factors include policy changes, household income and expenditure, housing characteristics, income poverty, and energy structure [106–108].

Foster, Tre, and Wodon [41] is one the earliest study to use the poverty line and consumption on energy expenditures to analyze energy

poverty. Later, Pachauri and Spreng [42] merged access and consumption to make a two-dimensional measure for energy poverty in India and found that energy poverty in India declined from 1983 to 2000. Energy poverty is also worsening in developed countries due to declining income and increasing energy prices [79]. Based on the methods proposed by Nussbaumer et al. [98], a recent study by Ogumike and Ozughalu [109], and Tabata and Tsai [110] found income poverty as the main factor influencing energy poverty.

One of the significant factors that drive energy poverty is poor energy efficiency in the heating/cooling of buildings. Using the Italian Household Budget Survey 2018, Betto et al. [111] provided empirical evidence for the association between energy poverty and building inefficiency. Furthermore, Karpinska and Śmiech [65] also documented that building inefficiency is the leading cause of energy poverty in Central and Eastern European countries. Similarly, Primc et al. [112] and Legendre and Ricci [113] found energy poverty is linked with the inefficiency of heating systems in buildings. Characteristics of the household head play a vital role in predicting the likelihood of energy poverty. Earlier studies reported that education, employment status, and gender of the household head are associated with the household's energy poverty status [112]. Likewise, Ogumike and Ozughalu [109] and Tabata and Tsai [110] provided empirical evidence for household head's education explaining the energy poverty status of the household. Fresh empirical evidence is bringing new factors that drive energy poverty. Recently financial inclusion has been found as a significant determinant of energy poverty in Turkey [94] and Ghana [47]. Fuel poverty was also found to be associated with subjective well-being for a large sample in Australia [30].

Another major factor affecting the energy poverty status of households reported in the literature is the number of household members or family size. Miniaci et al. [114] discussed the association between family size and the incidence of energy poverty. Their study found empirical evidence to show the linkage between energy poverty and family size for Italy. Dogan et al. [94] and Koomson and Danquah [47] also provided evidence for a significant association between household size and energy poverty.

4. Data sampling and methodology

4.1. Data & sampling

The present study uses data from eight waves of the Household Integrated Expenditure Survey (HIES) covering the period from 1998 to 2019 to examine the energy poverty situation in Pakistan and the factor influencing energy poverty. HIES is a nationally representative data set with detailed information about household expenditures on various goods and services, including fuel. Pakistan Bureau of Statistics conducted this survey using two-stage stratified random sampling techniques. The procedure includes a sampling framework for both rural and urban areas. The sampling framework of HIES is based on Census 1998 and the updated census of 2017. The universe for the survey consists of all four provinces of Pakistan.

A two-stage stratified sampling framework was adopted for this household survey. Primary sampling units (PSU) are selected for both rural and urban areas at the first stage. At the second stage, households were selected randomly from these PSUs. For rural areas, villages served as PSU, while in the urban area, PSU was further classified into enumeration blocks and divided into three groups based on income. Each enumeration block consists of 200–250 households in well-defined boundaries. A specified number of 16 households from both urban and rural sample PSU are selected with equal probability using a systematic sampling technique with a random start [62]. Therefore, the sample representative for both national and provincial levels for urban and rural settings. The distribution of sample size in each survey is given in Table 1.

Table 1

Distribution of the sample size by survey.

Year	Rural	Urban	Total households
HIES 1998-99	9184	5561	14,745
HIES 2001-02	9180	5597	14,777
HIES 2005-06	9212	6239	15,451
HIES 2007-08	9255	6253	15,508
HIES 2011-12	9063	6743	15,806
HIES 2013-14	11,755	6234	17,989
HIES 2015-16	16,155	8083	24,238
HIES 2018-19	15,936	8873	24,809
Total Sample	81,246	61,291	143,323

4.2. Empirical methods

Following [115,116], we used the Probit model for analysis as the dependent variable (energy poverty) is bivariate (i.e., 1 if the household is energy poor; 0 otherwise). The Probit model estimates the probability of occurrence of an event rather than the outcome itself.

$$\varphi^{-1}(p_i) = \sum_{k=0}^{k=n} \beta_k x_{ik} \quad (1)$$

where $p_i = 1$ if the household is energy poor, and 0 otherwise, according to the LIHC method, x_{ik} represents independent side variables. We use household characteristics, socio-demographic and economic variables as the independent-explanat or variables.

Further, from the probit model, we can also estimate marginal effects for each independent variable on the propensity of a household to be energy poor. This marginal effect reflects the impact of a unit change in each independent variable on the probability $P(Y = 1|X = x)$, keeping other variables constant. The following equation shows marginal effects estimation.

$$\partial P(y_i = 1|x_i) / \partial x_i = \partial E(y_i|x_i) / \partial x_i = \varphi(x_i'\beta) \quad (2)$$

To estimate the marginal effect for each independent variable (continuous) in the model, we follow

$$ME = \frac{1}{n} \sum_{i=1}^n \varphi(x_i'\beta) \quad (3)$$

To estimate the marginal effect for each independent variable (dummy) in the model, we follow

$$ME = \frac{1}{n} \sum_{i=1}^n [\varphi(x_i'\beta|x_i^k = 1) - \varphi(x_i'\beta|x_i^k = 0)] \quad (4)$$

As discussed in section 3.2, there are various methods available in the literature to measure energy poverty, the present study preferred the ten percent method and the double median method for three reasons. Firstly, these two are expenditure-based approaches, and to the best of our knowledge, there is no prior research on Pakistan using this approach. Studies such as [50,51,58,60] included non-expenditure-based methods to measure energy poverty in Pakistan. Secondly, in the context of the proposed energy sector reforms in Pakistan, including eliminating subsidies and increment in tariffs, analyzing the current expenditure burden on poor households is of great importance. Thirdly, comparison with other countries in the region is straightforward with these two approaches.

5. Result and discussion

5.1. Energy poverty in Pakistan

The present study attempts to estimate the prevalence of energy poverty in Pakistan using eight waves of HIES data spanning two decades from 1998 to 2019 and covering 142,537 households from urban

and rural Pakistan. Table 2 presents the percentage of households facing energy poverty based on ten percent and the double median method for each year. These estimates clearly show a gradual increase in energy poverty across Pakistan. The percentage of households experiencing energy poverty was 19.2% in 1999 using the ten percent method, which increased to 28.6% in 2019. However, as per the double median method, energy poverty increased from 7.6% in 1999 to 11.9% in 2019. The energy poverty estimates differ between different methods adopted [115]. Based on the estimates presented in Table 2, the Energy Poverty situation has worsened in the last 20 years, which is a cause of concern, particularly in the light of SDGs 7, which aims to provide clean and affordable energy for all by 2030. Therefore, a granular examination of the poverty situation and its determinants in Pakistan is crucial for initiating policy changes to improve the energy poverty situation in Pakistan.

Furthermore, Fig. 1 shows the percentage of energy-poor households across the income quartile. Energy poverty decreases across the income quartile irrespective of the methods adopted to estimate poverty. The poorest quartile has the highest rate of energy poverty, and it is the least in the richest quartile. According to the top ten percent approach, 35.6% of the poorest households were energy-poor, while only 15% were energy poor in the richest quartile. Similarly, as per the double median approach, the percentage of poor households is 16% among the poorest households while 5% among the richest households.

Fig. 2 shows the relationship between education level and prevalence of energy poverty. It shows that energy poverty is prevalent mainly among households whose head is illiterate. With the increase in the level of education of household heads, energy poverty declines sharply. As per the double median approach, about 28.8% of the households among the illiterate group were energy-poor, while only 17.3% of the households with more than ten years of education of household head were energy poor. As per the ten percent approach, about 11.4% of the households among the illiterate group were energy-poor, while only 5.8% of the households with more than ten years of education of household head were energy-poor.

Fig. 3 shows that female-headed households are more energy-poor than male-headed households, and the finding is consistent for two different energy poverty measures. As per the double median approach, 9.3% of the male-headed households were energy-poor, while 16% of the female-headed households were energy-poor.

Table 3 presents descriptive statistics of the variables used to estimate energy poverty. As expected, energy poverty was more prevalent in rural Pakistan. Ten percent methods show 27% of the rural household were energy-poor while only 19% of the urban households were energy-poor. The double median approach shows 10.7 and 7.3% of households in rural and urban households, respectively, were energy poor. The different results from the two approaches confirm the argument by Ntaintasis et al. [37]; Dogan et al. [94] that the rate of energy poverty varies across different methods used to estimate energy poverty.

As anticipated, the years of schooling of household heads is 3.5 and 6.6 years in rural and urban, respectively. As indicated by the assets index, rural households are poorer compared to urban households. Furthermore, the percentage of households that own agricultural land are 34 and 07 in rural and urban areas, respectively. Additionally, the

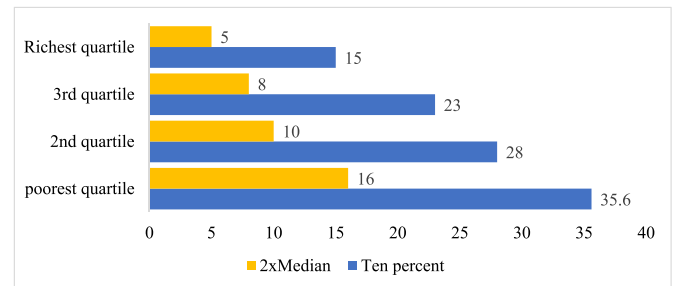


Fig. 1. Energy poverty across income quartile.

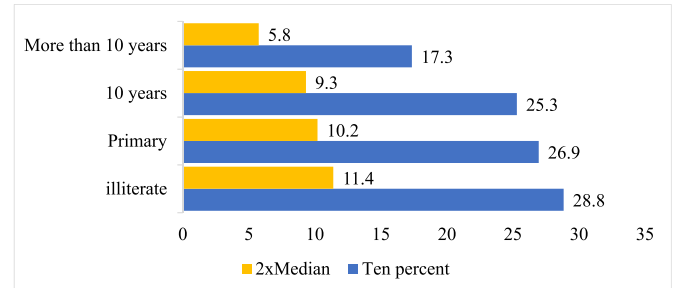


Fig. 2. Education level of head and distribution energy poverty.

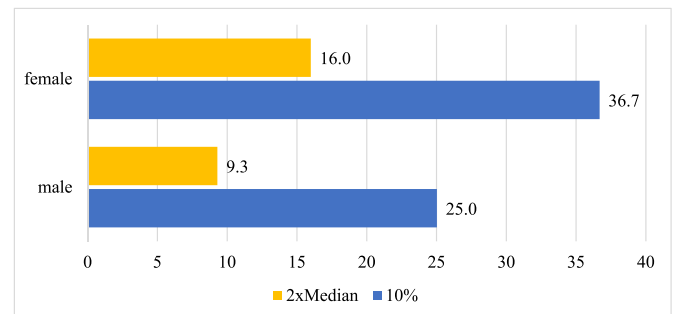


Fig. 3. Gender of head and distribution of energy poverty.

type of toilets was used as another proxy for wealth in the current study. Statistics show that households with a flush toilet connected to sewerage is 3 and 43% in rural and urban areas, respectively. Households without toilets stand at 33 in a rural area and 2.3% urban.

5.2. Factor influencing energy poverty

To find the determinants of energy poverty, we used pooled data from eight rounds of the HIES/PSLM survey (1998–2019). Following [115–118], we used the Probit model to explore factors influencing energy poverty in Pakistan, and the marginal effects are reported in Table 4. The probit model is suitable for binary variables, while other regression techniques such as ordinary least square (OLS) parameters are no longer efficient because of heteroskedastic regression errors [119]. The assumption of OLS is violated when using a dichotomous dependent variable [120]. In a Probit model, the dependent variable has two mutually exclusive categories. The dependent variable in the present study is energy poverty, which takes value of 1 if the household is energy poor and zero if non-energy poor. We estimated four probit models with different specifications. Specification 1 is based on ten percent methods with assets index and without the year 2015–16 because 2015–16 did not have assets variable while specification 2 is without assets index and with the year 2015–16. Specification 3 and 4 is based on double median methods-while specification 3 is with assets

Table 2
Energy poverty Trends of (percentage of energy-poor households by year).

Survey	10%	2xMedian
1999	19.2	07.6
2002	25.3	08.4
2006	26.7	07.9
2008	25.1	07.3
2012	25.8	08.3
2014	20.5	08.6
2016	23.1	11.1
2019	28.6	11.9

Table 3

Descriptive statistics (n = 143,012).

	Total (n = 142,537)	Rural (n = 81,246)	Urban (n = 61,129)
	Mean/% (S.D)	Mean/% (S.D)	Mean/% (S.D)
Energy poverty %			
Ten percent	24 (43)	27 (44)	19 (39)
Double median	9.2 (29)	10.7 (31)	7.3 (26)
Family size	6.67 (3.3)	6.83 (3.43)	6.52 (3.28)
Education of head in years	4.8 (5.03)	3.54 (4.47)	6.60 (5.20)
Household head female (%)	8.6	8.9	8.3
Asset (index by PCA)	3.4 (2.0)	-0.75 (1.4)	1.21 (2.201)
Log of the current value of livestock	2.4 (4.7)	3.96 (5.4)	0.43 (2.2)
Households with agriculture land (%)	22	34 (0.47)	7 (0.25)
Number of rooms	2.4 (1.4)	2.28 (1.39)	2.58 (1.45)
Toilet type %			
Flush with sewerage	20	3	43
Flush with pit	31	33	29
Open flush	13	9	20
Dry raised latrine	06	10	2
Dry pit latrine	07	11	2
No toilet	20	33	2.3
Occupancy type %			
House owned	85	92	74.5
Rented	08	2	17
Subsidized rent	1.7	0.04	3.5
Rent-free house	05	5	4.5
Employment status of head			
Head as non farm worker (%)	81	69.9	97.1

Note: Standard deviation values are provided in parenthesis for continuous variables only.

index and without the year 2015–16, specification 4 is without assets index, and with the year 2015–16 because 2015–16 did not have assets variable. Furthermore, we also tested the robustness of our estimates by employing logit model [121]. All the data cleaning and estimation exercises were performed using STATA 16.

The coefficient of the age of the household head is negative and significant on energy poverty in three out of four model specifications indicating that the households with the elderly head are less likely to be energy-poor. The negative impact of head's age on energy poverty is also reported in previous literature [63,94,115] for Ethiopia, Bangladesh, and Turkey, respectively. Rehmat and Mirza [51] also found similar results for Pakistan using PSLM data and measuring energy poverty by MEPI. However, the results contrast the findings of Koomson and Danquah [47] for Ghana and Awaworyi et al. [122] for Australia. With an increase in the age of the household head, families accumulate resources and social capital, which reduces them to falling into energy poverty.

As expected, the coefficient of the female-headed households dummy is positive and significant at a 1% level of significance in all four models. This finding confirms the results for Turkey obtained by Dogan et al. [94]. This implies that female-headed households are at high risk of facing energy poverty in Pakistan, requiring special attention in the energy policy.

The family size was negatively associated with energy poverty, which means that households with large family sizes are less likely to be energy poor. Our findings contradict the findings of Dogan et al. [94] for Turkey, Awaworyi et al. [122] for Australia, Mohr [123] for the USA, and Ismail and Khembo [124] for South Africa. However, for Ghana, Koomson, and Danquah [47], found a nonlinear relationship between energy poverty and family size. Therefore, it can be argued that for Turkey, Australia, and the USA, family size is positively related to energy poverty due to the small family size and low dependency ratio Legendre and Ricci [86], while in the case of Pakistan, it is negative due to large family size and high dependency. Similar to our study, Rehmat and Mirza [51] found a negative relationship between family size and

Table 4

Probit estimates of the determinants of energy poverty: Coefficients of marginal effects.

	(Model 1) Ten%	(Model 2) Ten%	(Model 3) 2xMedian	(Model 4) 2xMedian
Demographic				
Head age	-0.0003*** (0.0001)	-0.0004*** (0.00009)	-0.00009 (0.00006)	-0.0001*** (0.00005)
Female headed household ^{a,b}	0.0278*** (0.0046)	0.0220*** (0.0041)	0.0135*** (0.002)	0.0109*** (0.0025)
Human capital				
Family size	-0.0097*** (0.0004)	-0.0104*** (0.0004)	-0.0048*** (0.0002)	-0.0055*** (0.0002)
Head schooling years	-0.0031*** (0.0003)	-0.0048*** (0.0002)	-0.0013*** (0.0001)	-0.0021*** (0.0001)
Head off-farm worker ^{a,c}	0.0175*** (0.0039)	0.0217*** (0.0036)	0.0049** (0.0023)	0.0067*** (0.0022)
Wealth				
Own agriculture land ^{a,d}	-0.0079** (0.0035)	-0.0029 (0.0033)	-0.0010 (0.0020)	0.0010 (0.0020)
Dry pit latrine ^{a,e}	0.00792 (0.0059)	0.0030 (0.0053)	-0.0069** (0.0034)	-0.0090*** (0.0031)
Dry raised latrine ^{a,e}	0.0499*** (0.0059)	0.0304*** (0.0055)	0.0147*** (0.0036)	0.0074** (0.0034)
Open flush ^{a,e}	0.0551*** (0.0053)	0.0308*** (0.0047)	0.0167*** (0.0032)	0.0067** (0.0028)
Flush with pit ^{a,e}	0.0442*** (0.0040)	0.0221*** (0.0036)	0.0112*** (0.0023)	0.0016 (0.0021)
Flush with sewerage ^{a,e}	0.0205*** (0.0056)	-0.0083* (0.0047)	0.0075** (0.0034)	-0.0067** (0.0028)
House in ownership ^{a,f}	0.0179*** (0.0048)	0.0165*** (0.0042)	0.0078*** (0.0028)	0.0060** (0.0026)
Subsidized rent ^{a,f}	-0.0110 (0.0110)	-0.0093 (0.0095)	0.0006 (0.0068)	-0.0013 (0.0060)
Rent free house ^{a,f}	0.0291*** (0.0077)	0.0278*** (0.0068)	0.0164*** (0.0048)	0.0159*** (0.0044)
Number of rooms	-0.0005 (0.0011)	-0.0044*** (0.0009)	-0.0006 (0.0006)	-0.0022*** (0.0006)
Asset index	-0.0139*** (0.0009)	-	-0.0059*** (0.0005)	-
Log of livestock value	-0.0023*** (0.0003)	-0.0017*** (0.0003)	-0.0014*** (0.0001)	-0.0011*** (0.0001)
Rural vs Urban				
Rural ^{a,g,h}	0.0428*** (0.0034)	0.0697*** (0.0030)	0.0166*** (0.0019)	0.0293*** (0.0018)
Province				
Sindh ^{a,h}	-0.186*** (0.0035)	-0.166*** (0.0033)	-0.0868*** (0.0017)	-0.0827*** (0.0018)
KPK ^{a,h}	0.0733*** (0.0052)	0.0713*** (0.0048)	0.0117*** (0.0029)	0.0132*** (0.0028)
Punjab ^{a,h}	0.0463*** (0.0045)	0.0323*** (0.0042)	0.0033 (0.0026)	-0.0019 (0.0025)
2001-02 ^{a,i}	0.0691*** (0.0057)	0.0727*** (0.0057)	0.0072** (0.0032)	0.0095*** (0.0034)
2005-06 ^{a,i}	0.0879*** (0.0058)	0.0889*** (0.0058)	0.0040 (0.0031)	0.0048 (0.0033)
2007-08 ^{a,i}	0.0594*** (0.0056)	0.0588*** (0.0056)	-0.0055* (0.0029)	-0.0059* (0.0031)
2011-12 ^{a,i}	0.0762*** (0.0057)	0.0757*** (0.0057)	0.0085*** (0.0032)	0.0093*** (0.0034)
2013-14 ^{a,i}	0.0199*** (0.0052)	0.0165*** (0.0052)	0.0127*** (0.0032)	0.0124*** (0.0034)
2015-16 ^{a,i} #	-	0.0682*** (0.0052)	-	0.0493*** (0.0037)
2018-19 ^{a,i}	0.104*** (0.0054)	0.0950*** (0.0053)	0.0417*** (0.0034)	0.0404*** (0.0036)
Observations	118,088	142,326	118,088	142,326
Pseudo R ²	.078	.068	.071	.063

Standard errors are in parentheses.

***p < .01, **p < .05, *p < .1.

#Data for durable assets variable used in models 1 and 3 is not available for the year 2015–16.

^a Dummy variable.

^b Base category: male-headed household.

^c Base category: household head in a farm.

^d Base category: no agricultural land.

^e Base category: no toilet.

^f Base category: rented house.

^g Base category: rural household.

^h Base category: Balochistan.

ⁱ Base category: the year 1998.

energy poverty.

The marginal effect of the education of the head is negative and significant at a 1% level of significance, which shows that as the number of schooling years of household head's increases, the likelihood of the household being energy-poor decreases. This result is consistent with the previous study conducted in Ethiopia by Alem and Demeke [115]. High education levels influence households' energy efficiency-related decisions and keep them out of energy poverty [86].

Household head's occupation in the off-farm sector has a positive effect on the energy poverty status of the household. Lin and Zhao [125] found opposite results for China; however, their study uses clean fuel as an indicator for the energy poverty index. The results show a mixed effect of the toilet type on energy poverty is mixed. We conducted separate probit regression for rural and urban samples (see Appendix Table 1A), which indicates the crucial role of economic status in reducing the energy poverty. The effect of toilet type-flush connected to sewerage is negative in both urban and rural samples for energy poverty measures.

The effect of house ownership shows a positive and significant on energy poverty, indicating that households with their own house are more energy poor. Rehmat and Mirza [50] found similar results for rural areas of Pakistan. Households in rent-free occupancy are also more likely to suffer from energy poverty. It may be because the rent-free and subsidized occupancy have free electricity and/or gas or a lump sum bill. This result is opposite to the finding from France [86]. The durable asset index showed a negative and significant relationship with energy poverty, highlighting that more affluent households are less energy poor. Asset index includes home appliances – fan, air conditioner, refrigerator, washing machine, cooler, television, and freezer-is negatively linked with energy poverty. Our result on energy poverty and durable assets is consistent with previous literature, for instance, in India Khandker et al. [116]. Mohr [121] found a negative association between air conditioner presence in households and energy poverty in USA. As expected, the result shows that the households in rural settings are more energy-poor, which may be due to access as well as lack of affordability. Urban dwellers are at lower risk of affordability problems than rural ones. This result is consistent with a study on Ghana [47].

Besides rural and urban, we used provincial dummy – Baluchistan (base/reference category), Punjab Sindh, and Khyber Pakhtoon Khawa (KPK) to control spatial heterogeneity. Marginal effects from the ten percent method show that compared to Baluchistan, Punjab and Khyber Pakhtoon Khawa (KPK) are relatively energy-poor, while Sindh is less energy-poor. However, results from the double median approach are insignificant for Punjab. Heterogeneity in energy poverty across the provinces in Pakistan is due to high variation in population and income.

In our estimation, the time dummies for various years show an increase in energy poverty; however, the previous literature based on the MEPI approach shows that energy poverty is declining [60]. Therefore, the expenditure approach of measuring energy poverty provokes policymakers to reduce energy prices and improve energy efficiency in Pakistan, while energy poverty is highly prevalent.

5.3. Robustness analysis on factors influencing energy poverty

To check for the robustness of the result, we estimated the probit model for rural and urban areas separately using ten percent and double median approach of energy poverty measurement (see Appendix Table A1). The result shows that in rural and urban areas, with the increase in the level of education of the household head, family size, and assets index, the likelihood of being energy poor decreases. In contrast,

female-headed households are more likely to be energy-poor in rural as well as urban areas. Further, we estimated the probit model by years and found a similar result in both ten percent and double median methods (see Appendix Table A2 and A3). For instance, in both ten percent and double median methods and for all the years, education level of the household, family size, and assets index were inversely associated with energy poverty while the rural dummy and female head dummy were positively and significantly associated with energy poverty. Finally, we also estimated the logit model for both methods for all years and found similar results (see Appendix Table A4 and A5).

6. Conclusion

Due to rising energy prices in many countries, the expenditure on energy is increasing, resulting in large share of household income going to pay energy bills which makes them energy-poor [80]. Energy poverty is linked with SDGs of poverty eradication (goal 1), health and well-being (goal 3), education (goal 4), and reducing inequality (goal 10), and climate change (goal 13), among others [71]. Given the importance of energy poverty and its connection with various issues in developing countries, it is crucial to examine the state of energy poverty in Pakistan. Therefore, in this study, an attempt has been made to capture the image of energy poverty status in Pakistan.

Although clean energy usage and electricity access were studied by previous research, we filled the research gap by relying on an expenditure approach based energy poverty. To achieve the goal of the present study, we employed data from eight surveys of HIES that are nationally representative datasets for Pakistan and cover a considerable time span between 1998 and 2019. Energy poverty measured through two different methods, namely the ten percent and double median approach, showed that energy poverty increased in Pakistan over this period. In addition, to guide policymakers on determinants of energy poverty we conducted regression analysis using the probit model. The findings showed that energy poverty is likely to be present in a household with female heads, households in rural areas, households with low income and low education.

Therefore, to achieve sustainable development goals, policymakers in Pakistan should pay special attention to the dimensions in which energy-poor households are worse off. Future policies related to energy supply in Pakistan should observe the discrimination of energy supply to energy-poor households, particularly in rural and disadvantageous regions [126].

Therefore, to achieve SDG 7 (affordable and clean energy) by 2030, the energy policy should first increase access to clean energy and re-structure the tariff, keeping in view the lowest income group. Furthermore, a gender-sensitive subsidy and tariff regime would help reduce energy poverty among females and improve the well-being of female-headed households. The investment in energy efficiency will reduce the cost of power generation and the financial burden incurred by energy expenditures on poor households. Additionally, a home appliance with increased energy efficiency will also reduce energy poverty among poor households.

Future studies may consider including measures for obsolete technologies in home appliances that some poor households mainly use and cause overconsumption of electricity. Such households must be distinguished from others that are using energy-conserving home appliances. Furthermore, under users of clean energy, such as households with low electricity expenditures, must be analyzed in detail to know the reasons for energy poverty. If data on arrears on electricity and gas bills are available, a more comprehensive study could be conducted to understand the depth of financial burden due to energy poverty. Future studies may analyze the relationship between energy poverty and health problems in Pakistan depending on the data availability. Furthermore, out-of-pocket health expenditure and time spent fetching firewood for cooking fuel could be new dimensions of energy poverty impacts analysis in Pakistan. Pakistan has recently decided to provide subsidies on

food items to 20 million households^{2,3} additional energy subsidies to these poor households could benefit them in multiple ways.

While the present study is conducted, the data is available for pre-pandemic time, however, the severity of energy poverty is likely to rise after the world has faced COVID-19. Therefore, future studies may consider the comparison of the pandemic on energy poverty.

Author contribution statement

Ashar Awan: Led the conceptualization of the research idea, data cleaning, analysis, write up, editing and structuring, revision. **Faik Bilgili:** Contributed in conceptualization of the research idea, data cleaning, analysis, write up, editing and structuring, revision. **Dil Bahadur Rahut:** Contributed in conceptualization of the research idea, data cleaning, analysis, write up, editing and structuring, revision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We would like to acknowledge with thanks the support from Asian Development Bank Institute, Tokyo, Japan. The authors are grateful to Mr. Maqsood Sadiq for his useful feedback. This paper is based on the Ph.D. dissertation of Ashar Awan. The views, information, or opinions expressed in papers are those of the authors, and the usual disclaimer applies.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rser.2022.112157>.

References

- [1] Smil V. *Energy in world history*. Routledge; 2019.
- [2] IEA. *Data and statistics*. 2021.
- [3] Sadath AC, Acharya RH. Assessing the extent and intensity of energy poverty using Multidimensional Energy Poverty Index: empirical evidence from households in India. *Energy Pol* 2017;102:540–50. <https://doi.org/10.1016/j.enpol.2016.12.056>.
- [4] Vera IA, Langlois LM, Rogner H-H, Jalal AI, Toth FL. Indicators for sustainable energy development: an initiative by the international atomic energy agency. *Nat Resour Forum* 2005;29:274–83. Wiley Online Library.
- [5] Khandker SR, Barnes DF, Samad HA. Are the energy poor also income poor? Evidence from India. *Energy Pol* 2012;47:1–12. <https://doi.org/10.1016/j.enpol.2012.02.028>.
- [6] Acharya RH, Sadath AC. Energy poverty and economic development: household-level evidence from India. *Energy Build* 2019;183:785–91. <https://doi.org/10.1016/j.enbuild.2018.11.047>.
- [7] Barnes DF. *Electric power for rural growth: how electricity affects rural life in developing countries*. Routledge; 2019.
- [8] Swain AK. *India's next generation of electricity priorities*. 2019.
- [9] Kanagawa M, Nakata T. Socio-economic impacts of energy poverty alleviation in rural areas of developing countries. *Proc. 26th USAEE/IAEE North Am. Conf* 2006;24–7. Ann Arbor, MI, USA.
- [10] Aristondo O, Onaibia E. Inequality of energy poverty between groups in Spain. *Energy* 2018;153:431–42. <https://doi.org/10.1016/j.energy.2018.04.029>.
- [11] Day R, Walker G, Simcock N. Conceptualising energy use and energy poverty using a capabilities framework. *Energy Pol* 2016;93:255–64.
- [12] Hills J. *Getting the measure of fuel poverty: final report of the fuel poverty review*. 2012.
- [13] Lin B, Wang Y. Does energy poverty really exist in China? From the perspective of residential electricity consumption. *Energy Pol* 2020;143:111557. <https://doi.org/10.1016/j.enpol.2020.111557>.
- [14] Maithani PC, Gupta D. *Achieving universal energy access in India: challenges and the way forward*. SAGE Publications India; 2015.
- [15] Recalde M, Peralta A, Oliveras L, Tirado-Herrero S, Borrell C, Palència L, et al. Structural energy poverty vulnerability and excess winter mortality in the European Union: exploring the association between structural determinants and health. *Energy Pol* 2019;133:110869.
- [16] Baudu R, Charlier D, Legendre B. Fuel poverty and health: a panel data analysis. 2020.
- [17] Churchill SA, Smyth R. Energy poverty and health: panel data evidence from Australia. *Energy Econ* 2021;97:105219.
- [18] Thomson H, Bouzarovski S, Snell C. Rethinking the measurement of energy poverty in Europe: a critical analysis of indicators and data. *Indoor Built Environ* 2017;26:879–901.
- [19] Kahouli S. An economic approach to the study of the relationship between housing hazards and health: the case of residential fuel poverty in France. *Energy Econ* 2020;85:104592.
- [20] Gillard R, Snell C, Bevan M. Advancing an energy justice perspective of fuel poverty: household vulnerability and domestic retrofit policy in the United Kingdom. *Energy Res Social Sci* 2017;29:53–61.
- [21] Mottaleb KA, Rahut DB, Ali A. An exploration into the household energy choice and expenditure in Bangladesh. *Energy* 2017;135:767–76. <https://doi.org/10.1016/j.energy.2017.06.117>.
- [22] González-Eguino M. Energy poverty: an overview. *Renew Sustain Energy Rev* 2015;47:377–85. <https://doi.org/10.1016/j.rser.2015.03.013>.
- [23] Niethammer C, Alstone P. Expanding women's role in Africa's modern off-grid lighting market: enhancing profitability and improving lives. *Gend Dev* 2012;20:145–57.
- [24] Patange OS, Ramanathan N, Rehman IH, Tripathi SN, Misra A, Kar A, et al. Reductions in indoor black carbon concentrations from improved biomass stoves in rural India. *Environ Sci Technol* 2015;49:4749–56.
- [25] Bryden KM, Bruce N, Peck M. WHO Indoor air quality guidelines: household fuel combustion. Geneva World Heal Organ; 2015.
- [26] Reyes R, Schueftan A, Ruiz C, González AD. Controlling air pollution in a context of high energy poverty levels in southern Chile: clean air but colder houses? *Energy Pol* 2019;124:301–11. <https://doi.org/10.1016/j.enpol.2018.10.022>.
- [27] Sokolowski J, Lewandowski P, Kielcawska A, Bouzarovski S. A multidimensional index to measure energy poverty: the Polish case. *Energy Sources B Energy Econ Plann* 2020;15:92–112.
- [28] Kyprianou I, Serghides D. Dealing with energy poverty in Cyprus—an overview. *Int J Sustain Energy* 2020;39:308–20.
- [29] Dagoumas A, Kitsios F. Assessing the impact of the economic crisis on energy poverty in Greece. *Sustain Cities Soc* 2014;13:267–78.
- [30] Churchill SA, Smyth R, Farrell L. Fuel poverty and subjective well-being. *Energy Econ* 2020;86:104650.
- [31] Thomson H, Snell C, Bouzarovski S. Health, well-being and energy poverty in Europe: a comparative study of 32 European countries. *Int J Environ Res Publ Health* 2017;14. <https://doi.org/10.3390/ijerph14060584>.
- [32] International Energy Agency. reportWEO-2017 special report: energy access outlook. <https://www.iea.org/Energyaccess/2017:1-143>. <https://doi.org/10.1016/j.ceo.2017.10.024> [pii].
- [33] Zhang D, Li J, Han P. A multidimensional measure of energy poverty in China and its impacts on health: an empirical study based on the China family panel studies. *Energy Pol* 2019;131:72–81.
- [34] Rahut DB, Das S, De Groote H, Behera B. Determinants of household energy use in Bhutan. *Energy* 2014;69:661–72. <https://doi.org/10.1016/j.energy.2014.03.062>.
- [35] Rahut DB, Behera B, Ali A. Household energy choice and consumption intensity: empirical evidence from Bhutan. *Renew Sustain Energy Rev* 2016;53:993–1009. <https://doi.org/10.1016/j.rser.2015.09.019>.
- [36] Bradshaw J, Hutton S. Social policy options and fuel poverty. *J Econ Psychol* 1983;3:249–66.
- [37] Ntaintasis E, Mirasgedis S, Tourkolias C. Comparing different methodological approaches for measuring energy poverty: evidence from a survey in the region of Attika, Greece. *Energy Pol* 2019;125:160–9. <https://doi.org/10.1016/j.enpol.2018.10.048>.
- [38] Sen A, Sen MA, Amartya S, Foster JE, Foster JE. *On economic inequality*. Oxford university press; 1997.
- [39] Wang Q, Kwan M, Fan J, Lin J. Racial disparities in energy poverty in the United States. *Renew Sustain Energy Rev* 2021;137:110620. <https://doi.org/10.1016/j.rser.2020.110620>.
- [40] Adusah-Poku F, Takeuchi K. Energy poverty in Ghana: any progress so far? *Renew Sustain Energy Rev* 2019;112:853–64.
- [41] Foster V, Tre J-P, Wodon Q. *Energy prices, energy efficiency, and fuel poverty*. Lat Am Caribb Reg Stud Program Washington, DC World Bank; 2000.
- [42] Pachauri S, Spreng D. Energy use and energy access in relation to poverty. *Econ Polit Wkly* 2004;271–8.
- [43] Miniaci R, Scarpa C, Valbonesi P. Fuel poverty and the energy benefits system: the Italian case. 2014. Available SSRN 2433441.
- [44] Wang K, Wang YX, Li K, Wei YM. Energy poverty in China: an index based comprehensive evaluation. *Renew Sustain Energy Rev* 2015;47:308–23. <https://doi.org/10.1016/j.rser.2015.03.041>.
- [45] Walter Boltz B, Irschik P, Pichler F, Jaquelin Cochran B, Bird L, Heeter J, et al. Getting it right: defining and fighting energy poverty in Austria. *ICER Chron* 2014;41–51.
- [46] González-Eguino M. Energy poverty: an overview. *Renew Sustain Energy Rev* 2015;47:377–85.

² <https://tribune.com.pk/story/2321388/govt-to-provide-cash-subsidies-to-marginalised-segments-tarin>.

³ <https://www.thenews.com.pk/print/910815-ehsaas-rashan-subsidy>.

- [47] Koomson I, Danquah M. Financial inclusion and energy poverty: empirical evidence from Ghana. *Energy Econ* 2021;94:105085. <https://doi.org/10.1016/j.eneco.2020.105085>.
- [48] Gouveia JP, Palma P, Simoes SG. Energy poverty vulnerability index: a multidimensional tool to identify hotspots for local action. *Energy Rep* 2019;5: 187–201. <https://doi.org/10.1016/j.egyr.2018.12.004>.
- [49] Heindl P, Schüssler R. Dynamic properties of energy affordability measures. *Energy Pol* 2015;86:123–32.
- [50] Rehmat A, Mirza FM. Multidimensional energy poverty in Pakistan : empirical evidence from household level micro data, vol. 155. Springer Netherlands; 2021. <https://doi.org/10.1007/s11205-020-02601-7>.
- [51] Rehmat A, Mirza FM. Determinants of multidimensional energy poverty in Pakistan : a household level analysis. Springer Netherlands; 2020. <https://doi.org/10.1007/s10668-020-01174-2>.
- [52] Swain A. Taking energy efficiency to the market. *Bus Stand* 2013;7.
- [53] UNDP. Annual report Pakistan SDGs. 2019.
- [54] United Nations T. Transforming our world: the 2030 agenda for sustainable development. 2015.
- [55] Swain AK. In a state of energy poverty. *The Hindu*; 2018. <https://www.thehindu.com/opinion/op-ed/in-a-state-of-energy-poverty/article23857192.ece>.
- [56] Birol F. Energy economics: a place for energy poverty in the agenda? *Energy J* 2007;28.
- [57] Mirza B, Kemp R. Why the rural rich remain energy poor. *Consilience* 2011: 133–56.
- [58] Sher F, Abbas A, Awan RU. An investigation of multidimensional energy poverty in Pakistan: a province level analysis. *Int J Energy Econ Pol* 2014;4:65.
- [59] Nussbaumer P, Nerini FF, Onyeji I, Howells M. Global insights based on the multidimensional energy poverty index (MEPI). *Sustain Times* 2013;5:2060–76. <https://doi.org/10.3390/su5052060>.
- [60] Mahmood R, Shah A. Deprivation counts : an assessment of energy poverty in Pakistan. *J The Lahore* 2017;1:109–32.
- [61] Haleem A, Khan MI, Athar F. Energy poverty and its impact ON environment. *CASE STUD PAKISTAN* 2020;1:68–89.
- [62] Nawaz S. Energy poverty, climate shocks, and health deprivations. *Energy Econ* 2021;100:105338. <https://doi.org/10.1016/j.eneco.2021.105338>.
- [63] Barnes DF, Khandker SR, Samad HA. Energy poverty in rural Bangladesh. *Energy Pol* 2011;39:894–904. <https://doi.org/10.1016/j.enpol.2010.11.014>.
- [64] Selçuk İŞ, Gölçek AG, Köktaş AM. Energy poverty in Turkey. *Sosyoekonomi* 2019; 27:283–99.
- [65] Karpinska L, Śmiech S. Conceptualising housing costs: the hidden face of energy poverty in Poland. *Energy Pol* 2020;147:111819.
- [66] Son H, Yoon S. Reducing energy poverty: characteristics of household electricity use in Vietnam. *Energy Sustain Dev* 2020;59:62–70. <https://doi.org/10.1016/j.esd.2020.08.007>.
- [67] Okushima S. Gauging energy poverty: a multidimensional approach. *Energy* 2017;137:1159–66. <https://doi.org/10.1016/j.energy.2017.05.137>.
- [68] Papada L, Kaliampakos D. Measuring energy poverty in Greece. *Energy Pol* 2016; 94:157–65. <https://doi.org/10.1016/j.enpol.2016.04.004>.
- [69] Pachauri S, Mueller A, Kemmler A, Spreng D. On measuring energy poverty in Indian households. *World Dev* 2004;32:2083–104.
- [70] Maryon-Davis A, Ballard T. Fuel poverty tackling cold homes and ill-health. 2014. London UK Heal. Forum.
- [71] Halkos GE, Gkampoura E-C. Coping with energy poverty: measurements, drivers, impacts, and solutions. *Energies* 2021;14. <https://doi.org/10.3390/en14102807>.
- [72] IEA. World energy outlook. Policy model and energy use 2002:21. 2002.
- [73] Winkler H, Simões AF, Rovere EL Ia, Alam M, Rahman A, Mwakasonda S. Access and affordability of electricity in developing countries. *World Dev* 2011;39: 1037–50. <https://doi.org/10.1016/j.worlddev.2010.02.021>.
- [74] Hills J. Fuel poverty: the problem and its measurement. Interim report of the Fuel Poverty Review. Centre for Analysis of Social Exclusion, LSE; 2011.
- [75] EUEO. Energy poverty and vulnerable consumers in the energy sector across the EU: analysis of policies and measures. INSIGHT E. Retrieved from EU Energy Observatory; 2015.
- [76] Sovacool BK, Cooper C, Bazilian M, Johnson K, Zoppo D, Clarke S, et al. What moves and works: broadening the consideration of energy poverty. *Energy Pol* 2012;42:715–9.
- [77] Sen A. Poverty: an ordinal approach to measurement. *Econ J Econ Soc* 1976: 219–31.
- [78] Sen A. Issues in the measurement of poverty. *Meas. Public choice*. Springer; 1981. p. 144–66.
- [79] Okushima S. Measuring energy poverty in Japan, 2004–2013. *Energy Pol* 2016; 98:557–64. <https://doi.org/10.1016/j.enpol.2016.09.005>.
- [80] Farrell L, Fry JM. Australia's gambling epidemic and energy poverty. *Energy Econ* 2021;97:105218. <https://doi.org/10.1016/j.eneco.2021.105218>.
- [81] Boardman B. Fuel poverty: from cold homes to affordable warmth. Pinter Pub Limited; 1991.
- [82] EBRD. Can the poor pay for power? The affordability of electricity in southeast Europe. 2003. Available Online Ww Ebrd Com/Country/Sector/Energyef/about/Powersee Pdf.
- [83] Heindl P. Measuring fuel poverty: general considerations and application to German household data. *Finanzarchiv* 2015;71:178–215. <https://doi.org/10.1628/001522115X14285723527593>.
- [84] Romero JC, Linares P, Otero XL, Labandeira X, Alonso AP. Pobreza Energética en España-Análisis económico y propuestas de actuación. Vigo: Econ Energy; 2014.
- [85] Imbert I, Nogues P, Sevenet M. Same but different: on the applicability of fuel poverty indicators across countries—insights from France. *Energy Res Social Sci* 2016;15:75–85.
- [86] Legendre B, Ricci O. Measuring fuel poverty in France: which households are the most fuel vulnerable? *Energy Econ* 2014;49:620–8. <https://doi.org/10.1016/j.eneco.2015.01.022>.
- [87] Roberts D, Vera-Toscana E, Phimister E. Fuel poverty in the UK: is there a difference between rural and urban areas? *Energy Pol* 2015;87:216–23.
- [88] Healy JD, Clinch JP. Quantifying the severity of fuel poverty, its relationship with poor housing and reasons for non-investment in energy-saving measures in Ireland. *Energy Pol* 2004;32:207–20. [https://doi.org/10.1016/S0301-4215\(02\)00265-3](https://doi.org/10.1016/S0301-4215(02)00265-3).
- [89] Townsend P. Poverty in the United Kingdom: a survey of household resources and standards of living. Univ of California Press; 1979.
- [90] Isherwood BC, Hancock RM. Household expenditure on fuel: distributional aspects', economic advisers office. London, United Kingdom: DHSS; 1979.
- [91] Kiely BE, Alam M, Blinman P, Tattersall MHN, Stockler MR. Estimating typical, best-case and worst-case life expectancy scenarios for patients starting chemotherapy for advanced non-small-cell lung cancer: a systematic review of contemporary randomized trials. *Lung Cancer* 2012;77:537–44.
- [92] Isaacman S, Becker R, Cáceres R, Kobourov S, Martonosi M, Rowland J, et al. Ranges of human mobility in Los Angeles and New York. In: 2011 IEEE int. Conf. Pervasive comput. Commun. Work. (PERCOM work. IEEE; 2011. p. 88–93.
- [93] McCarthy J, Peach RW. kAre home prices the next iBubblej?, I federal reserve bank of New York economic policy review. December 2004;1:17.
- [94] Dogan E, Madaleno M, Taskin D. Which households are more energy vulnerable ? Energy poverty and fi nancial inclusion in Turkey. *Energy Econ* 2021;99:105306. <https://doi.org/10.1016/j.eneco.2021.105306>.
- [95] Liddell C, Morris C, McKenzie SJP, Rae G. Measuring and monitoring fuel poverty in the UK: national and regional perspectives. *Energy Pol* 2012;49:27–32. <https://doi.org/10.1016/j.enpol.2012.02.029>.
- [96] Sovacool BK, Mukherjee I, Drupady IM, D'Agostino AL. Evaluating energy security performance from 1990 to 2010 for eighteen countries. *Energy* 2011;36: 5846–53. <https://doi.org/10.1016/j.energy.2011.08.040>.
- [97] Sovacool BK, Mukherjee I. Conceptualizing and measuring energy security: a synthesized approach. *Energy* 2011;36:5343–55. <https://doi.org/10.1016/j.energy.2011.06.043>.
- [98] Nussbaumer P, Bazilian M, Modi V. Measuring energy poverty: focusing on what matters. *Renew Sustain Energy Rev* 2012;16:231–43. <https://doi.org/10.1016/j.rser.2011.07.150>.
- [99] Moore R. Definitions of fuel poverty: implications for policy. *Energy Pol* 2012;49: 19–26.
- [100] Bradshaw J, Middleton S, Davis A, Oldfield N, Smith N, Cusworth L, et al. A minimum income standard for Britain: what people think. 2008.
- [101] Rademaekers K, Yearwood J, Ferreira A, Pye S, Hamilton I, Agnolucci P, et al. Setting indicators to measure energy poverty. Rotterdam, The Netherlands: Trinomics; 2016.
- [102] Castaño-rosa R, Solís-guzmán J, Rubio-bellido C, Marrero M. Towards a multiple-indicator approach to energy poverty in the European Union : a review. *Energy Build* 2019;193:36–48. <https://doi.org/10.1016/j.enbuild.2019.03.039>.
- [103] Adzawla W, Alhassan H, Jongare AI. Explaining the effects of socioeconomic and housing characteristics on the choice of toilet facilities among Ghanaian households. *J Environ Publ Health* 2020;2020.
- [104] Belaid F. Exposure and risk to fuel poverty in France: examining the extent of the fuel precariousness and its salient determinants. *Energy Pol* 2018;114:189–200.
- [105] Robinson C, Bouzarovski S, Lindley S. 'Getting the measure of fuel poverty': the geography of fuel poverty indicators in England. *Energy Res Social Sci* 2018;36: 79–93.
- [106] Crensil AO, Asuman D, Fenny AP. Assessing the determinants and drivers of multidimensional energy poverty in Ghana. *Energy Pol* 2019;133:110884.
- [107] Mendoza Jr CB, Cayonte DDD, Leabres MS, Manaligod LRA. Understanding multidimensional energy poverty in the Philippines. *Energy Pol* 2019;133: 110886.
- [108] Sharma SV, Han P, Sharma VK. Socio-economic determinants of energy poverty amongst Indian households: a case study of Mumbai. *Energy Pol* 2019;132: 1184–90.
- [109] Ogwumike FO, Ozughalu UM. Analysis of energy poverty and its implications for sustainable development in Nigeria. *Environ Dev Econ* 2016;21:273–90.
- [110] Tabata T, Tsai P. Fuel poverty in Summer: an empirical analysis using microdata for Japan. *Sci Total Environ* 2020;703:135038. <https://doi.org/10.1016/j.scitotenv.2019.135038>.
- [111] Betto F, Garengo P, Lorenzoni A. A new measure of Italian hidden energy poverty. *Energy Pol* 2020;138:111237.
- [112] Primc K, Slabe-Erker R, Majcen B. Constructing energy poverty profiles for an effective energy policy. *Energy Pol* 2019;128:727–34.
- [113] Legendre B, Ricci O. Measuring fuel poverty in France : which households are the most fuel vulnerable. *Energy Econ* 2015;49:620–8. <https://doi.org/10.1016/j.eneco.2015.01.022>.
- [114] Miniaci R, Scarpa C, Valbonesi P. Energy affordability and the benefits system in Italy. *Energy Pol* 2014;75:289–300. <https://doi.org/10.1016/j.enpol.2014.09.008>.
- [115] Alem Y, Demeke E. The persistence of energy poverty : a dynamic probit analysis. *Energy Econ* 2020;90:104789. <https://doi.org/10.1016/j.eneco.2020.104789>.
- [116] Khandker SR, Barnes DF, Samad HA. Are the energy poor also income poor? Evidence from India. *Energy Pol* 2012;47:1–12. <https://doi.org/10.1016/j.enpol.2012.02.028>.

- [117] Brown H, Vera-Toscano E. Energy poverty and its relationship with health: empirical evidence on the dynamics of energy poverty and poor health in Australia. *SN Bus Econ* 2021;1:139. <https://doi.org/10.1007/s43546-021-00149-3>.
- [118] Oum S. Energy poverty in the Lao PDR and its impacts on education and health. *Energy Pol* 2019;132:247–53. <https://doi.org/10.1016/j.enpol.2019.05.030>.
- [119] Stone M, Rasp J. Tradeoffs in the choice between logit and OLS for accounting choice studies. *Account Rev* 1991:170–87.
- [120] Noreen E. An empirical comparison of probit and OLS regression hypothesis tests. *J Account Res* 1988:119–33.
- [121] Prakash K, Munyanyi ME. Energy poverty and obesity. *Energy Econ* 2021;101: 105428. <https://doi.org/10.1016/j.eneco.2021.105428>.
- [122] Awaworyi Churchill S, Smyth R. Ethnic diversity, energy poverty and the mediating role of trust: evidence from household panel data for Australia. We thank two referees for constructive comments. This article uses unit record data from the Household, Income and Labour Dynamics in Austr. *Energy Econ* 2020; 86:104663. <https://doi.org/10.1016/j.eneco.2020.104663>.
- [123] Mohr TM. Fuel poverty in the US: evidence using the 2009 residential energy consumption survey. *Energy Econ* 2018;74:360–9. <https://doi.org/10.1016/j.eneco.2018.06.007>.
- [124] Ismail Z, Khembo P. Determinants of energy poverty in South Africa. *J Energy South Afr* 2015;26:66–78.
- [125] Lin B, Zhao H. Does off-farm work reduce energy poverty? Evidence from rural China. *Sustain Prod Consum* 2021;27:1822–9. <https://doi.org/10.1016/j.spc.2021.04.023>.
- [126] Swain AK, Dubash NK. Beyond Poles and wires: how to keep the electrons flowing?. 2019.
- [127] Jäntti M, Danziger Sbt-H of ID. Chapter 6 Income poverty in advanced countries, vol. 1. Elsevier; 2000. p. 309–78. [https://doi.org/10.1016/S1574-0056\(00\)80009-3](https://doi.org/10.1016/S1574-0056(00)80009-3).